Exercises: Day 4

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Deductive Arguments

- **Deductive arguments** are arguments that aim to draw a conclusion from premises via logical consequence

- A valid argument is a deductive argument such that if the premises are true then so is the conclusion

  1. If John is a human being then John is a mammal
  2. John is a human being
  3. Hence, John is a mammal

- There is no way for 1 and 2 to be true and 3 to be false

Inductive Examples

(1) There are 20 oranges in this bag
(2) 12 oranges have been inspected and are delicious
(3) Hence, **most** of the other 8 oranges in the bag will be delicious

(1) There are 20 oranges in this bag
(2) This randomly chosen orange is delicious
(3) Hence, **most** of the remaining 19 oranges will be delicious

Abductive Arguments

1. There are multiple signals of troop movement near a contested border.
2. Each of the following explain 1:
   - H_1: Routine military exercises.
   - H_2: Preparation for an invasion.
   - H_3: Defensive training mission.
3. There has been increased political tension in the area.
4. Previous intelligence indicates potential military escalation.
5. The best explanation given 1, 3, and 4 is H_2.

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AGAIN, THE CONCLUSION IS NOT GUARANTEED TO FOLLOW

Zebra Puzzle

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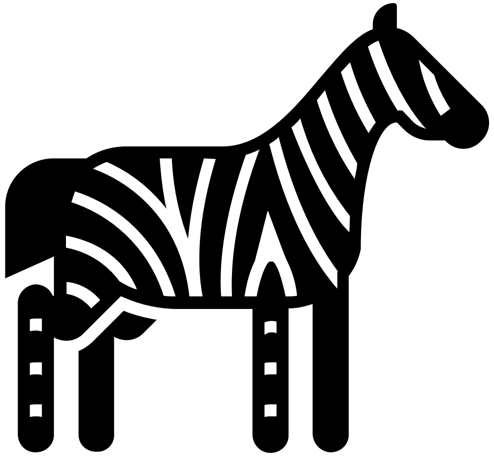

WHO OWNS THE ZEBRA?

Goals

- Highlight the difference between **deductive** and **abductive reasoning**
- Gain practice encoding ontology commitments in OWL to support deductive reasoning
- Gain practice leveraging generative AI and OWL encodings to support abductive reasoning
- Evaluate the outputs of automated reasoners vs generative AI

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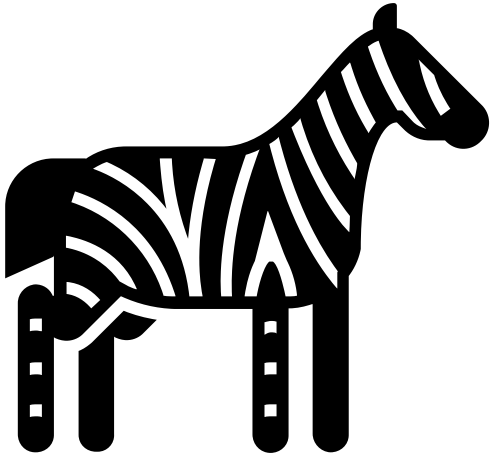

WHO OWNS THE ZEBRA?

Styles of Automated Reasoning

- Logicians have for decades leveraged automated theorem provers and model checkers to explore logical space

- Most run by leveraging **resolution**, or **unification**, or **tableau algorithms**

Prover9

```
============================== PROOF ==========

% -------- Comments from original proof -------
% Proof 1 at 0.00 (+ 0.03) seconds.
ngth of proof is 7.
vel of proof is 3.
ximum clause weight is 2.
ven clauses 2.

-> Q # label(non_clause).   [assumption].
# label(non_clause) # label(goal).   [goal].
3 -P | Q.   [clausify(1)].
4 P.   [assumption].
5 -Q.   [deny(2)].
6 Q.   [ur(3,a,4,a)].
7 $F.   [resolve(6,a,5,a)].

============================== end of proof ===
```

Practical Reasoning for Expressive Description Logics

Ian Horrocks[1], Ulrike Sattler[2], and Stephan Tobies[2]

[1] Department of Computer Science, University of Manchester[†]
[2] LuFG Theoretical Computer Science, RWTH Aachen[‡]

Abstract. Description Logics (DLs) are a family of knowledge representation formalisms mainly characterised by constructors to build complex concepts and roles from atomic ones. Expressive role constructors are important in many applications, but can be computationally problematical. We present an algorithm that decides satisfiability of the DL $\mathcal{ALC}$ extended with transitive and inverse roles, role hierarchies, and qualify-

HermiT OWL Reasoner

The New Kid on the OWL Block

Analytic Tableau

Construct a tree where each branch represents an interpretation of a proposition, decompose proposition by extended branches with new interpretations on each, close branch if reach a contradiction, terminate when all branches closed or one full branch open

Tableau

- **Conjunction Rule**: If given (C_1 & C_2) on branch b then add C_1 and C_2 to b
- **Disjunction Rule**: If given (C_1 v C_2) on branch b_1 then add C_1 to new branch b_1 and C_2 to new branch b_2
- **Existential Rule**: If ∀x∃y r(x,y) & Cy on branch b_1 then add r(x,a) and Ca to a new branch, as long as "a" fresh
- **Universal Rule**: ∀x∀y r(x,y) → Cy & r(x,y) on b then add Cy to b

Tableau

⊓-rule
- **Conjunction Rule**: If given (C_1 & C_2) on branch b then add C_1 and C_2 to b

⊔-rule
- **Disjunction Rule**: If given (C_1 v C_2) on branch b_1 then add C_1 to new branch b_1 and C_2 to new branch b_2

∃-rule
- **Existential Rule**: If ∀x∃y r(x,y) & Cy on branch b_1 then add r(x,a) and Ca to a new branch, as long as "a" fresh

∀-rule
- **Universal Rule**: ∀x∀y r(x,y) → Cy & r(x,y) on b then add Cy to b

Method of Analytic Tableau

- Is the following set of propositions **consistent**?

$$\exists y\ s(a,x)\ \&\ F(x)$$

$$s(a,b)$$

$$\forall y\ s(a,y) \rightarrow \neg F(y) \vee \neg B(y)$$

$$B(b)$$

Tableau Algorithm

- If so, each can be decomposed into a tree in which one branch is open

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rdfs:comment [language: en]

Note 1: Each house is painted exactly one color; each house is painted a different color.
Note 2: Each house has exactly one human occupant of distinct nationality and exactly one distinct pet is owned by that human.
Note 3: Each human occupant drinks exactly one distinct beverage and smokes exactly one distinct brand of cigarettes.

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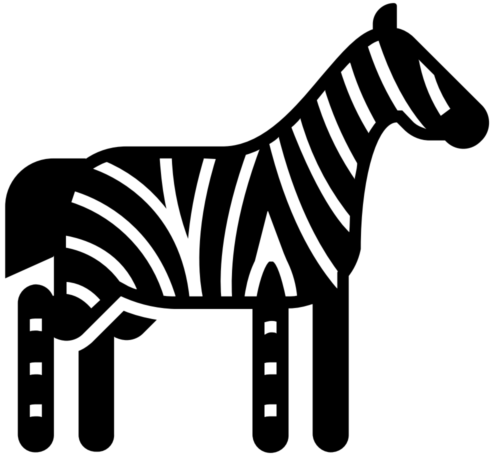

WHO OWNS THE ZEBRA?

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House

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lives_in
owned_by
owns

drinks
drunk_by

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- house_2
- house_3
- house_4
- house_5

- red

- green

- tea
- ukrainian

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zebra !

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Favorite Model

- By all means pick your favorite model and prompt it to make an abductive guess as to who owns the zebra
- **Caveat:** You should tell the model not to leverage online resources, this is a widely known puzzle and the solution is available
- That would undermine the task, we want to see how good the models are at abductive reasoning
- Pure examples of this experiment create **novel puzzles to test**

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Add more axioms to your model and run the experiment again!

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Run HermiT in Protege and see what follows.
Feed your agent and ask it to guess the answer.

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- Beverage
- Cigarette
- Color
- House
- Man
- Pet

Zebra Puzzle

Classes allow you to represent uncertainty; you know that there are five men, you are trying to uncover instance-level facts about them

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- home_of
- left_of
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- owned_by
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- right_of
- smoked_by
- smokes

Zebra Puzzle

We do this by representing logical relationships between instances of classes using object properties

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Zebra Puzzle

Inverse Of

has_color

If R is the inverse of S, then for any <x,y> in R, <y,x> is in S

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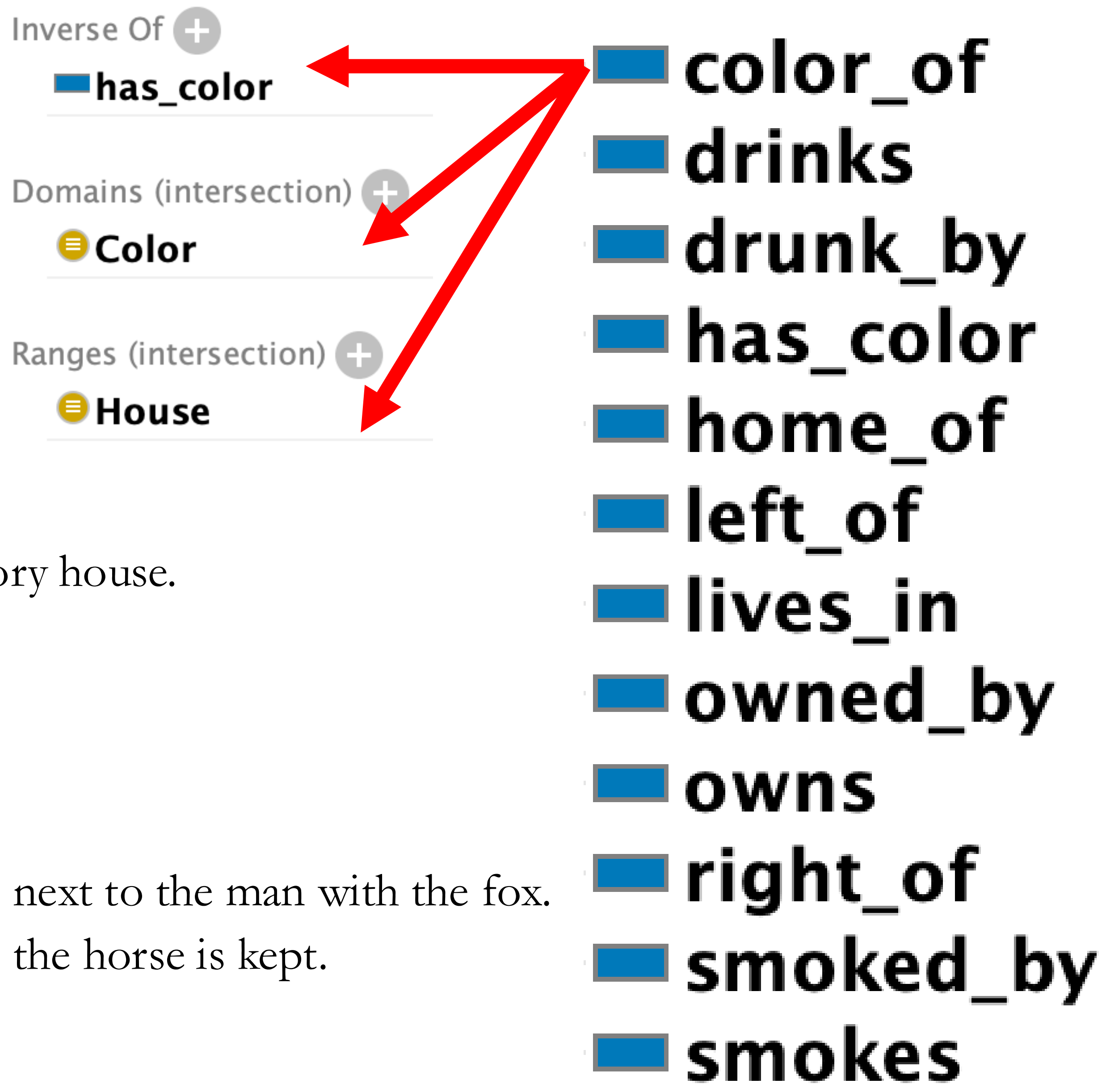

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Inverse Of

has_color

Domains (intersection)

Color

Ranges (intersection)

House

For color_of(x,y), x must be a color

- color_of
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Inverse Of

has_color

Domains (intersection)

Color

Ranges (intersection)

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☑ Functional

☑ Inverse functional

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For color_of(x,y), there is a 1-1 relationship between color and house

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- blue
- chesterfields
- coffee
- dog
- englishman
- fox
- green
- horse
- house_1
- house_2
- house_3
- house_4
- house_5
- ivory
- japanese
- kools
- lucky_strikes
- milk
- norwegian
- old_gold
- orange_juice
- parliaments
- red
- snail
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- yellow
- zebra

Zebra Puzzle

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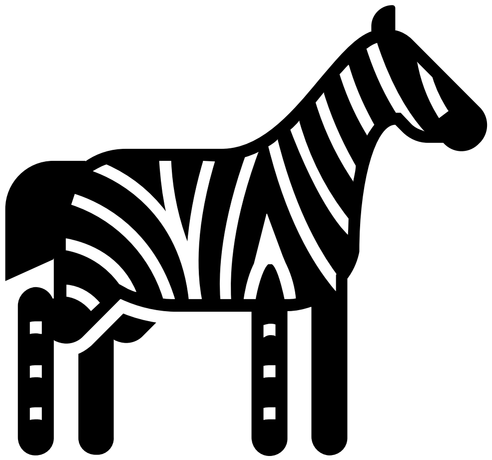

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Zebra Puzzle

Color

color_of some (left_of some (has_color value ivory))

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Zebra Puzzle

color_of some (right_of some (has_color value ivory))

green is the color of some x

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Zebra Puzzle

color_of **some** (right_of **some** (has_color **value** ivory))

...and since the domain of color_of is colors and the range is houses, it follows that x is a house

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Zebra Puzzle

color_of **some** (right_of **some** (has_color **value** ivory))

Where that house is to the right of some x

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Inverse Of
left_of

Domains (intersection)
House

Ranges (intersection)
House

☑ Functional
☑ Inverse functional

- color_of
- drinks
- drunk_by
- has_color
- home_of
- left_of
- lives_in
- owned_by
- owns
- right_of
- smoked_by
- smokes

Zebra Puzzle

color_of some (right_of some (has_color value ivory))

Where that house is to the right of some house

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Zebra Puzzle

color_of some (right_of some (has_color value ivory))

...that is ivory

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Altogether, green is the color of some house x that is to the right of some house y that has color ivory

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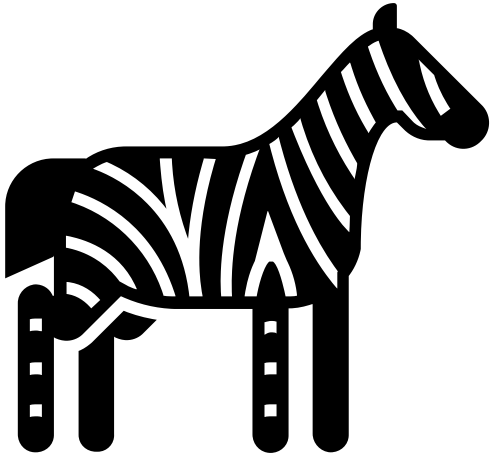

WHO OWNS THE ZEBRA?

OWL vs Abductive Reasoning

- You are able to deductively prove who owns the zebra if you’ve done this exercise correctly
- Run HermiT in Protege and see what follows
- Feed your agent and ask it to guess the answer

OWL vs Abductive Reasoning

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YOU SHOULD GET THE SAME ANSWER…